

Code to Reproduce Figures and Tables in “A Matched Study Design to Estimate the Effect of Furosemide in Horse Racing”

Michele Sezgin

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)  
library(here)
```

here() starts at /Users/michelesezgin/Desktop/projects/TAS-horseracing

```
library(brms)
```

Loading required package: Rcpp

Loading 'brms' package (version 2.23.0). Useful instructions can be found by typing `help('brms')`. A more detailed introduction to the package is available through `vignette('brms_overview')`.

Attaching package: 'brms'

The following object is masked from 'package:stats':

ar

```
library(tidyr)
library(survival)
```

Attaching package: 'survival'

The following object is masked from 'package:brms':

kidney

```
library(tidyr)
library(latex2exp)
library(scales)
library(forcats)

set.seed(2025)
theme_set(theme_bw())

source(here('data', 'data-loaders', 'load_data.R'))
```

Section 3.1 - Exploratory Data Analysis

Figure 1

a

```
# avg number of with and without
avg_counts <- data_clean |>
  group_by(registration_number, med_description) |>
  summarize(n = n()) |>
  ungroup() |>
  group_by(med_description) |>
  summarize(avg = mean(n) |>
```

```

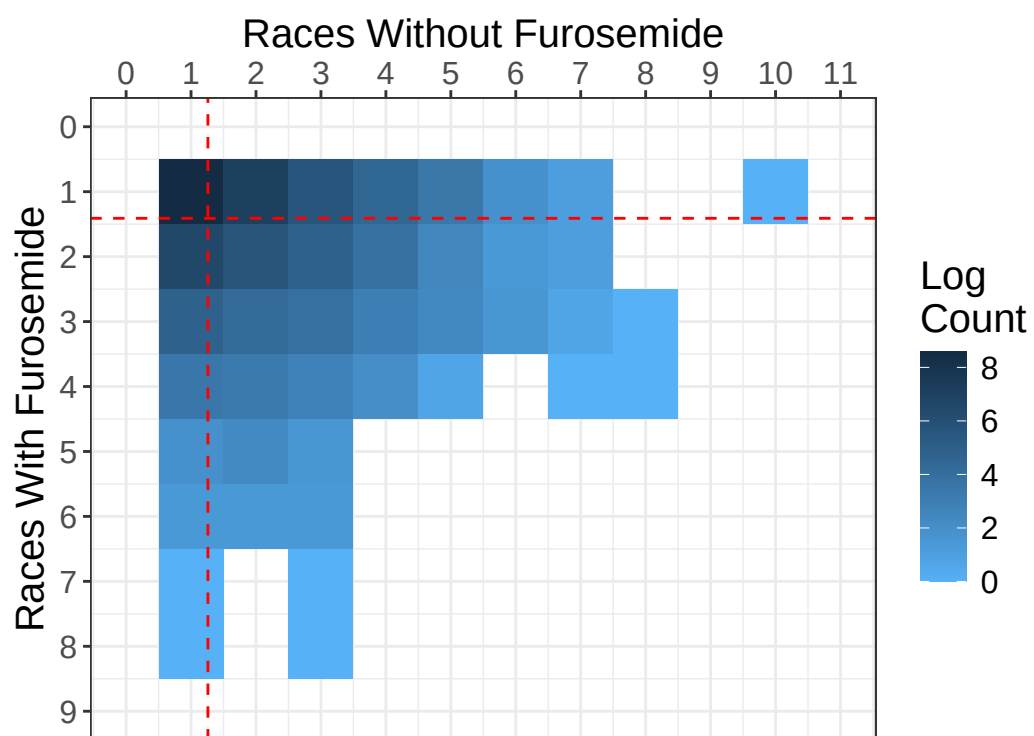
        round(2))

avg_no_lasix <- avg_counts |>
  filter(med_description == 'No Medication') |>
  pull(avg)

avg_lasix <- avg_counts |>
  filter(med_description == 'Lasix') |>
  pull(avg)

data_clean |>
  group_by(registration_number, lasix_ind) |>
  summarize(num_races = n()) |>
  ungroup() |>
  pivot_wider(names_from = lasix_ind, values_from = num_races) |>
  rename(lasix = `1`, no_lasix = `0`) |>
  group_by(lasix, no_lasix) |>
  summarize(count = n(), log_count = log(n())) |>
  ggplot(aes(x = lasix, y = no_lasix, fill = log_count)) +
  geom_tile() +
  labs(x = "Races Without Furosemide",
       y = "Races With Furosemide",
       fill = "Log\nCount") +
  coord_fixed() +
  scale_y_discrete() +
  scale_y_reverse(limits = c(9, 0), breaks = 0:9) +
  scale_x_continuous(position = "top",
                     limits = c(0, 11),
                     breaks = 0:11) +
  scale_fill_continuous(high = "#132B43", low = "#56B1F7") +
  geom_hline(linetype = 'dashed',
             yintercept = avg_lasix,
             color = 'red') +
  geom_vline(linetype = 'dashed',
             xintercept = avg_no_lasix,
             color = 'red') +
  theme(text = element_text(size = 15))

```

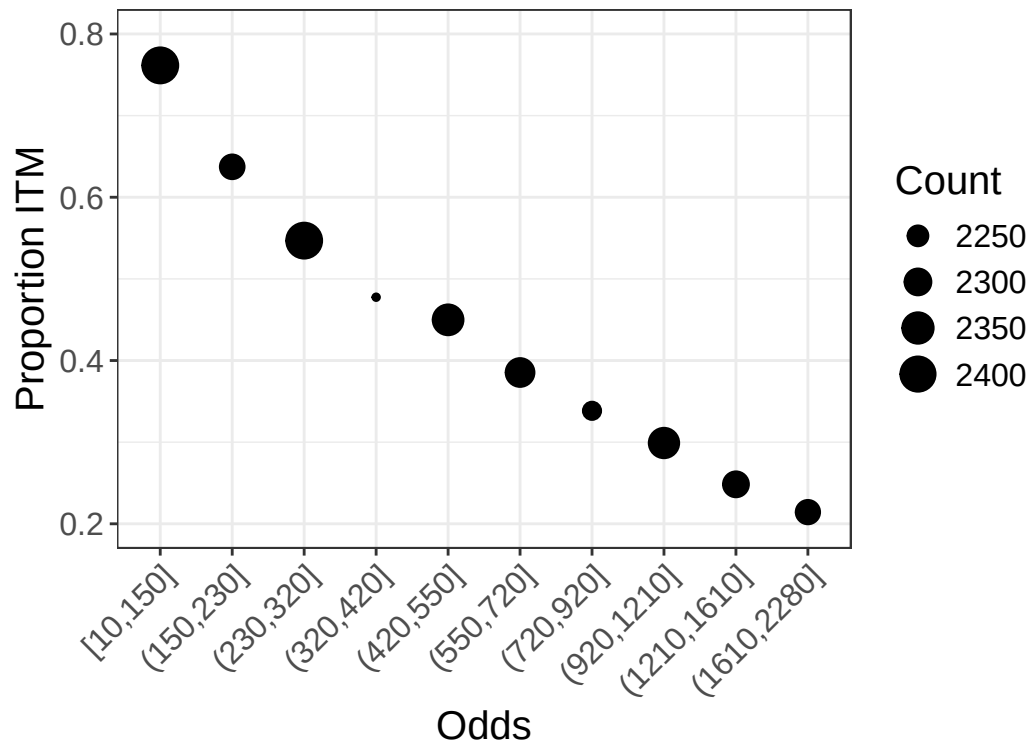


b

```

data_clean |>
  mutate(odds_grouped = cut_number(odds, n = 10, dig.lab = 10)) |>
  group_by(odds_grouped) |>
  summarize(
    avg_itm = mean(in_the_money),
    logit_itm = rje::logit(avg_itm),
    count = n()
  ) |>
  ggplot(aes(x = odds_grouped, y = avg_itm)) +
  geom_point(aes(size = count)) +
  theme(axis.text.x = element_text(
    angle = 45,
    vjust = 1,
    hjust = 1
  )) +
  labs(x = 'Odds', y = 'Proportion ITM', size = 'Count') +
  theme(text = element_text(size = 15)) +
  scale_y_continuous(limits = c(0.2, 0.8))

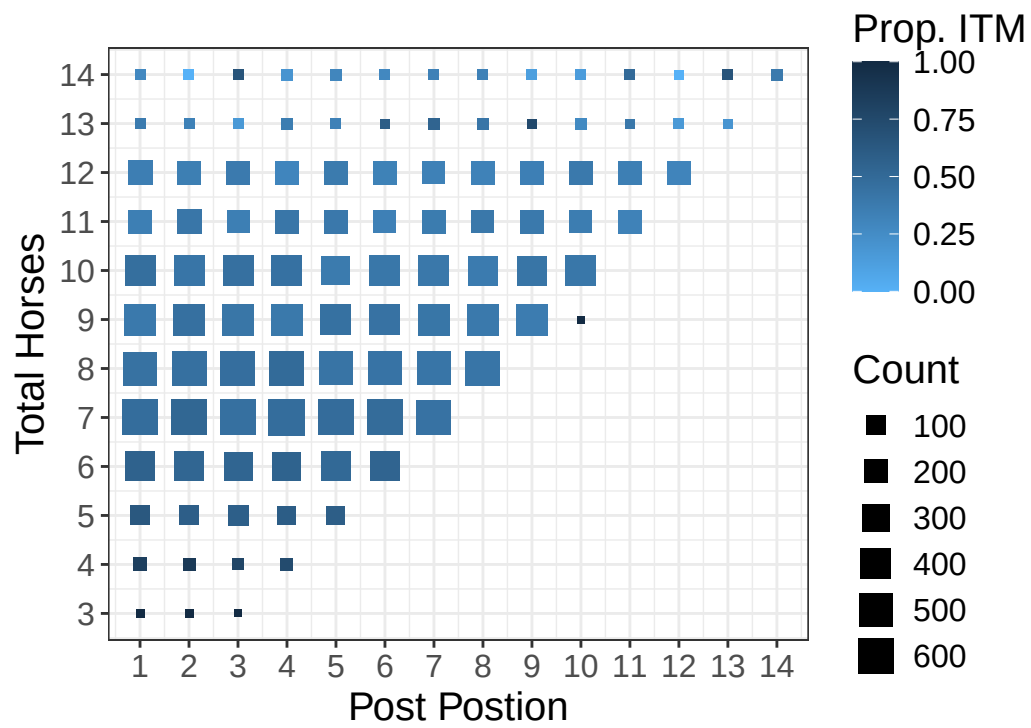
```



c

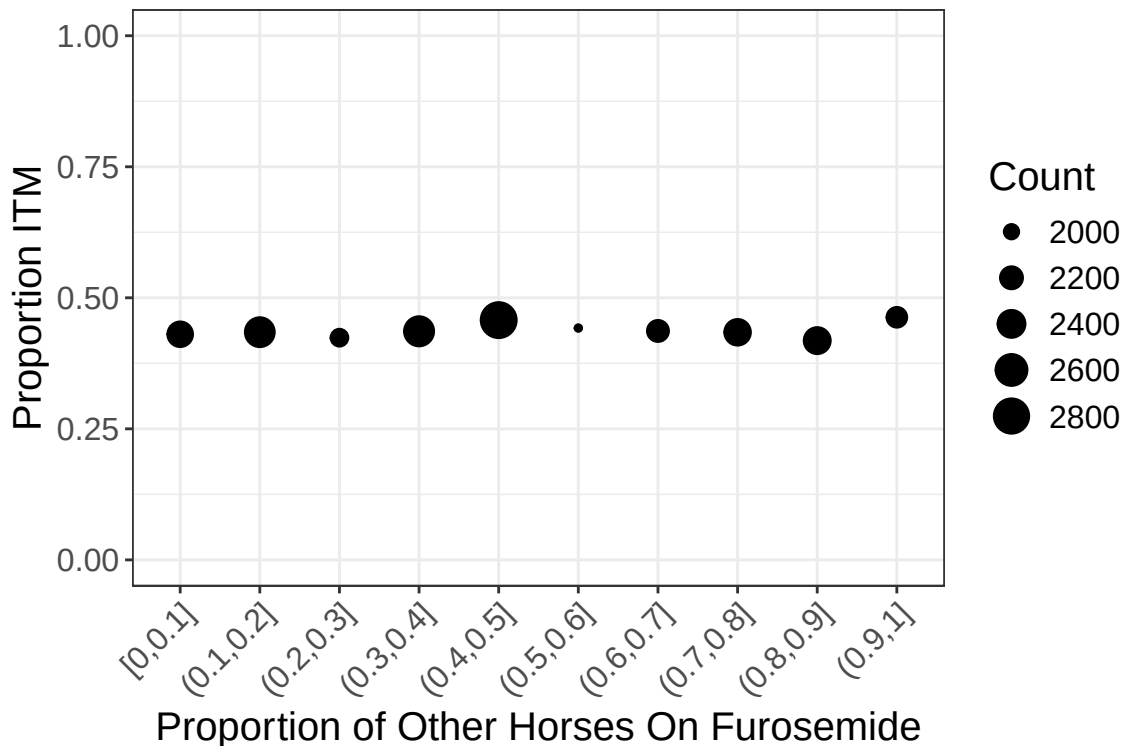
```
pp_data <- data_clean |>
  mutate(total_horses = total_other_horses + 1) |>
  group_by(post_position, total_horses) |>
  summarize(avg_itm = mean(in_the_money), n = n())

pp_data |>
  ggplot(aes(
    x = post_position,
    y = total_horses,
    color = avg_itm,
    size = n
  )) +
  geom_point(shape = 'square') +
  coord_fixed() +
  scale_color_gradient(high = "#132B43", low = "#56B1F7") +
  labs(x = 'Post Position',
       y = 'Total Horses',
       size = 'Count',
       color = 'Prop. ITM') +
  theme(text = element_text(size = 15)) +
  scale_y_continuous(limits = c(3, 14), breaks = 3:14) +
  scale_x_continuous(limits = c(1, 14), breaks = 1:14)
```



d

```
data_clean |>
  mutate(frac_lasix = frac_lasix * .01,
         fl_gr = cut_interval(frac_lasix, n = 10)) |>
  group_by(fl_gr) |>
  summarize(avg_itm = mean(in_the_money), n = n()) |>
  ggplot(aes(x = fl_gr, y = avg_itm, size = n)) +
  geom_point() +
  ylim(0, 1) +
  theme(axis.text.x = element_text(
    angle = 45,
    vjust = 1,
    hjust = 1
  )) +
  labs(x = 'Proportion of Other Horses On Furosemide',
       y = 'Proportion ITM',
       size = 'Count') +
  theme(text = element_text(size = 15))
```



Section 3.2.1 - Conditional Logistic Regression

Table 1

```
# 3.2.1 conditional logistic regressison
clogit_fit <- readRDS(here('data', 'models', 'clogit_fit.rds'))
# 3.2.2 BHLR
brms_fit_apples <- readRDS(here('data', 'models', 'brms_fit_apples.rds'))
# A.1 BHLR random slopes
# (not used really, just here for cross-referencing divergent transition +
# variance credible interval nearly containing zero claims)
brms_fit_rs <- readRDS(here('data', 'models', 'brms_fit_rs.rds'))
# A.1 BHLR
brms_fit <- readRDS(here('data', 'models', 'brms_fit.rds'))
```

These are the outputs necessary for reproducing Table 1.

```
# 3.2.1 conditional logistic
summary(clogit_fit)
```

Call:

```
coxph(formula = Surv(rep(1, 23109L), in_the_money) ~ med_description +
      post_position + splines::ns(odds, df = 4) + frac_lasix +
      total_other_horses + strata(registration_number), data = data_clean,
      method = "exact")
```

n= 23109, number of events= 10120

	coef	exp(coef)	se(coef)	z	Pr(> z)
med_descriptionLasix	0.630753	1.879025	0.034816	18.117	<2e-16 ***
post_position	-0.018738	0.981436	0.007324	-2.559	0.0105 *
splines::ns(odds, df = 4)1	-1.134111	0.321708	0.110147	-10.296	<2e-16 ***
splines::ns(odds, df = 4)2	-1.282123	0.277448	0.122325	-10.481	<2e-16 ***
splines::ns(odds, df = 4)3	-2.369992	0.093482	0.252060	-9.402	<2e-16 ***
splines::ns(odds, df = 4)4	-1.201334	0.300793	0.130394	-9.213	<2e-16 ***
frac_lasix	-0.009679	0.990368	0.001061	-9.120	<2e-16 ***
total_other_horses	-0.114051	0.892213	0.012515	-9.113	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

exp(coef) exp(-coef) lower .95 upper .95

med_descriptionLasix	1.87903	0.5322	1.75508	2.0117
post_position	0.98144	1.0189	0.96745	0.9956
splines::ns(odds, df = 4)1	0.32171	3.1084	0.25924	0.3992
splines::ns(odds, df = 4)2	0.27745	3.6043	0.21830	0.3526
splines::ns(odds, df = 4)3	0.09348	10.6973	0.05704	0.1532
splines::ns(odds, df = 4)4	0.30079	3.3245	0.23296	0.3884
frac_lasix	0.99037	1.0097	0.98831	0.9924
total_other_horses	0.89221	1.1208	0.87059	0.9144

Concordance= 0.663 (se = 0.007)

Likelihood ratio test= 1118 on 8 df, p=<2e-16

Wald test = 941.6 on 8 df, p=<2e-16

Score (logrank) test = 1056 on 8 df, p=<2e-16

```
exp(confint(clogit_fit))
```

	2.5 %	97.5 %
med_descriptionLasix	1.75508114	2.0117221
post_position	0.96744956	0.9956253
splines::ns(odds, df = 4)1	0.25924157	0.3992263
splines::ns(odds, df = 4)2	0.21830224	0.3526176
splines::ns(odds, df = 4)3	0.05703897	0.1532074
splines::ns(odds, df = 4)4	0.23295704	0.3883820
frac_lasix	0.98831024	0.9924303
total_other_horses	0.87059428	0.9143680

```
# 3.2.2 BHLR model
```

```
summary(brms_fit_apples)
```

Loading required namespace: rstan

```
Family: bernoulli
Links: mu = logit
Formula: in_the_money ~ med_description + (1 | registration_number) + splines::ns(odds, df =
Data: data_clean (Number of observations: 23109)
Draws: 4 chains, each with iter = 8000; warmup = 4000; thin = 1;
total post-warmup draws = 16000
```

Multilevel Hyperparameters:

```
~registration_number (Number of levels: 8644)
```

```
Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
```

sd(Intercept)	0.29	0.06	0.15	0.40	1.00	1579	1382
---------------	------	------	------	------	------	------	------

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	2.20	0.10	2.00	2.40	1.00	24244
med_descriptionLasix	0.40	0.03	0.34	0.46	1.00	26040
splines::nsoddsdfEQ41	-2.08	0.08	-2.24	-1.92	1.00	22380
splines::nsoddsdfEQ42	-2.18	0.09	-2.36	-2.00	1.00	27667
splines::nsoddsdfEQ43	-4.36	0.19	-4.73	-4.01	1.00	23415
splines::nsoddsdfEQ44	-2.13	0.10	-2.32	-1.94	1.00	30800
post_position	-0.02	0.01	-0.03	-0.01	1.00	35604
frac_lasix	-0.00	0.00	-0.00	0.00	1.00	16154
total_other_horses	-0.08	0.01	-0.10	-0.07	1.00	27352

	Tail_ESS
Intercept	14333
med_descriptionLasix	11721
splines::nsoddsdfEQ41	13878
splines::nsoddsdfEQ42	12211
splines::nsoddsdfEQ43	12561
splines::nsoddsdfEQ44	12677
post_position	12826
frac_lasix	9461
total_other_horses	12743

Draws were sampled using `sample(hmc)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat` = 1).

```
fixef(brms_fit_apples, probs = c(0.025, 0.975)) |>
  exp()
```

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	9.00317325	1.107973	7.394988196	11.00866566
med_descriptionLasix	1.49130820	1.030503	1.407143482	1.58246325
splines::nsoddsdfEQ41	0.12497185	1.083828	0.106565016	0.14641664
splines::nsoddsdfEQ42	0.11285708	1.097334	0.094117173	0.13530727
splines::nsoddsdfEQ43	0.01273194	1.205517	0.008792708	0.01820116
splines::nsoddsdfEQ44	0.11903967	1.103435	0.098179975	0.14404090
post_position	0.98238092	1.005651	0.971586587	0.99311998
frac_lasix	0.99949773	1.000509	0.998504543	1.00050370
total_other_horses	0.91919408	1.008326	0.904248967	0.93417692

Section 3.2.2 - BHLR

```
brms_pred <- readRDS(here('data', 'datasets', 'brms_preds.rds'))
brms_pred_apples <- readRDS(here('data', 'datasets', 'brms_preds_apples.rds'))

calibration_data <- data.frame(
  clogit = predict(clogit_fit, type = 'lp', reference = 'strata') |>
    rje::expit(),
  brms = brms_pred[, 1],
  brms_apples = brms_pred_apples[, 1],
  actual = data_clean$in_the_money
) |>
  tidyr::pivot_longer(c(clogit, brms, brms_apples),
    names_to = 'model',
    values_to = 'pred') |>
  mutate(model = case_when(
    model == 'clogit' ~ 'Cond. Logit',
    model == 'brms_apples' ~ 'BHLR',
    # this is the BHLR fit with extra random effects
    # terms from A.1
    T ~ 'Model 3'
  ))
```

Figure 2

```
calibration_data |>
  filter(model == 'BHLR') |>
  mutate(interval = cut(
    pred,
    breaks = seq(0, 1, by = .1),
    include.lowest = T
  )) |>
  group_by(interval) |>
  summarize(
    int_pred = mean(pred),
    int_actual = mean(actual),
    se = sqrt((int_actual * (1 - int_actual)) / n()),
    n = n(),
    ci_lower = pmax(int_actual - 2 * se, 0),
```

```

    ci_upper = pmin(int_actual + 2 * se, 1)
  ) |>
  ggplot(aes(x = int_pred, y = int_actual)) +
  geom_point(alpha = 0.5, aes(size = n)) +
  geom_errorbar(aes(ymin = ci_lower, ymax = ci_upper)) +
  geom_smooth(se = FALSE,
             method = 'loess',
             color = 'black') +
  geom_abline(slope = 1,
             intercept = 0,
             linetype = "dashed") +
  labs(x = "Predicted probability",
       y = "Observed fraction",
       color = 'Model',
       size = 'Count') +
  ylim(0, 1) +
  theme(text = element_text(size = 15))

```

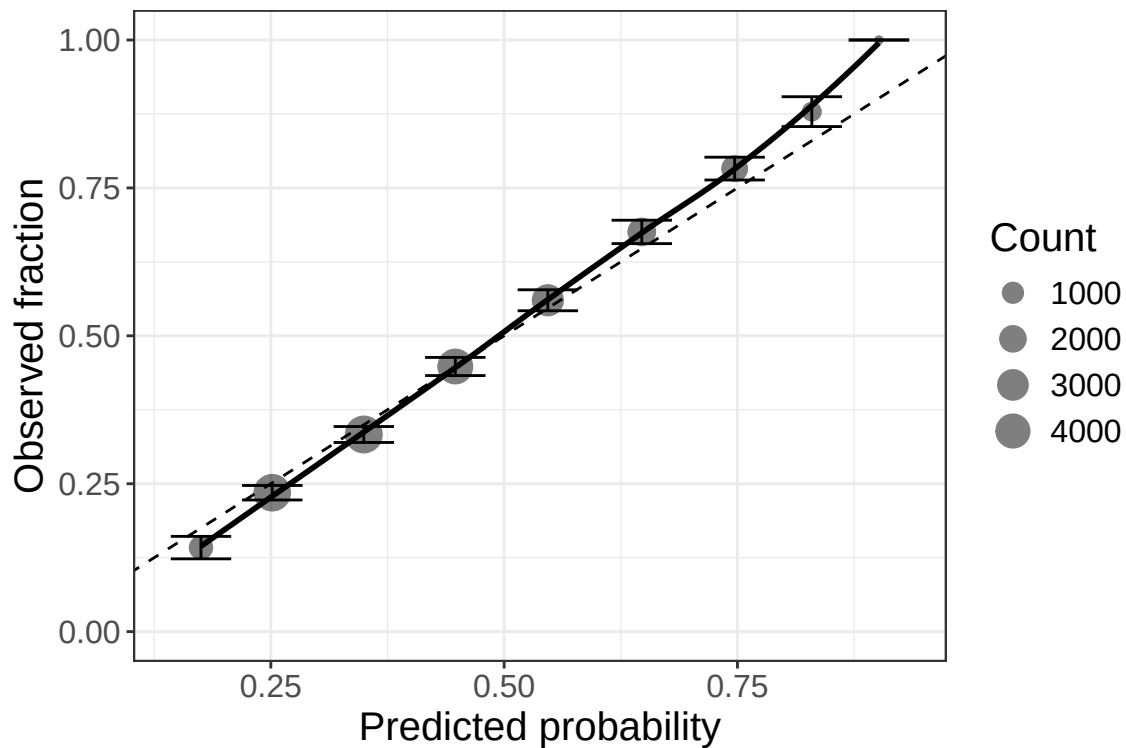


Figure 1: Predicted probabilities binned into 0.1-wide intervals.

Section A.1.1 Results

This is not used to produce any figures or tables but shows the results of fitting the random slopes model described in section A.1:

```
# BHLR random slopes model described in sec A.1
summary(brms_fit_rs)
```

Warning: There were 8 divergent transitions after warmup. Increasing adapt_delta above 0.95 may help. See <http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup>

```
Family: bernoulli
Links: mu = logit
Formula: in_the_money ~ med_description + (med_description | registration_number) + (1 | track_id)
Data: data_clean (Number of observations: 23109)
Draws: 4 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 16000
```

Multilevel Hyperparameters:

~jockey_id (Number of levels: 1935)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.11	0.04	0.02	0.18	1.00	1955	2302

~registration_number (Number of levels: 8644)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
sd(Intercept)	0.27	0.09	0.05	0.44	1.01
sd(med_descriptionLasix)	0.14	0.10	0.01	0.37	1.00
cor(Intercept,med_descriptionLasix)	-0.15	0.56	-0.96	0.92	1.00

	Bulk_ESS	Tail_ESS
sd(Intercept)	506	401
sd(med_descriptionLasix)	730	1288
cor(Intercept,med_descriptionLasix)	1688	3535

~track_id (Number of levels: 88)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.13	0.03	0.08	0.19	1.00	5681	7437

~track_id:post_position (Number of levels: 808)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.08	0.04	0.01	0.16	1.00	1799	3402

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	2.24	0.11	2.03	2.46	1.00	12881
med_descriptionLasix	0.40	0.03	0.34	0.46	1.00	10895
splines::nsoddsdfEQ41	-2.09	0.08	-2.26	-1.94	1.00	11748
splines::nsoddsdfEQ42	-2.20	0.09	-2.38	-2.02	1.00	12200
splines::nsoddsdfEQ43	-4.39	0.19	-4.77	-4.02	1.00	10633
splines::nsoddsdfEQ44	-2.14	0.10	-2.34	-1.95	1.00	15046
frac_lasix	-0.00	0.00	-0.00	0.00	1.00	16812
total_other_horses	-0.10	0.01	-0.11	-0.08	1.00	16931
	Tail_ESS					
Intercept	12903					
med_descriptionLasix	10434					
splines::nsoddsdfEQ41	12325					
splines::nsoddsdfEQ42	11701					
splines::nsoddsdfEQ43	11312					
splines::nsoddsdfEQ44	12366					
frac_lasix	10688					
total_other_horses	11615					

Draws were sampled using `sample(hmc)`. For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Figure A.1

```
# extract the data necessary to make trace plots: samples and iterations.
brms_apples_data <- brms_fit_apples$fit@sim$samples |>
  purrr::list_rbind(names_to = 'index') |>
  group_by(index) |>
  mutate(iter = 1:n()) |>
  ungroup()

brms_data <- brms_fit$fit@sim$samples |>
  purrr::list_rbind(names_to = 'index') |>
  group_by(index) |>
  mutate(iter = 1:n()) |>
  ungroup()
```

```

# forest plots
lower <- brms_apples_data |>
  dplyr::select(
    'b_med_descriptionLasix',
    'b_frac_lasix',
    'b_total_other_horses',
    'b_post_position'
  ) |>
  sapply(quantile, probs = .025) |>
  sapply(exp) |>
  unname()

upper <- brms_apples_data |>
  dplyr::select(
    'b_med_descriptionLasix',
    'b_frac_lasix',
    'b_total_other_horses',
    'b_post_position'
  ) |>
  sapply(quantile, probs = .975) |>
  sapply(exp) |>
  unname()

means <- brms_apples_data |>
  dplyr::select(
    'b_med_descriptionLasix',
    'b_frac_lasix',
    'b_total_other_horses',
    'b_post_position'
  ) |>
  colMeans() |>
  sapply(exp) |>
  unname()

# names <- brms_apples_data |>
#   dplyr::select(starts_with('b')) |>
#   colnames()

names <- fct_inorder(c(
  'Furosemide',
  'Frac. Furosemide',
  'Total Other Horses',

```

```

    'Post Position'
  ))

brms_apples_forest_data <- tibble(
  coef = names,
  mean = means,
  lower = lower,
  upper = upper
)

lower <- brms_data |>
  dplyr::select('b_med_descriptionLasix',
                'b_frac_lasix',
                'b_total_other_horses') |>
  sapply(quantile, probs = .025) |>
  sapply(exp) |>
  unname()

upper <- brms_data |>
  dplyr::select('b_med_descriptionLasix',
                'b_frac_lasix',
                'b_total_other_horses') |>
  sapply(quantile, probs = .975) |>
  sapply(exp) |>
  unname()

means <- brms_data |>
  dplyr::select('b_med_descriptionLasix',
                'b_frac_lasix',
                'b_total_other_horses') |>
  colMeans() |>
  sapply(exp) |>
  unname()

names <- fct_inorder(c('Furosemide',
                       'Frac. Furosemide',
                       'Total Other Horses'))

brms_forest_data <- tibble(
  coef = names,
  mean = means,
  lower = lower,

```

```

    upper = upper,
    model = 'Model 3'
  )

combined_data <- brms_apples_forest_data |>
  mutate(model = 'Model 2') |>
  bind_rows(brms_forest_data)

ggplot(data = combined_data) +
  geom_vline(
    xintercept = 1,
    linetype = 'dashed',
    color = 'red',
    linewidth = .75,
    alpha = .5
  ) +
  geom_errorbar(
    aes(
      xmin = lower,
      xmax = upper,
      y = fct_rev(coef),
      color = model
    ),
    linewidth = .75,
    alpha = .5
  ) +
  geom_point(
    shape = 21,
    fill = 'white',
    aes(x = mean, y = fct_rev(coef), color = model),
    size = 3,
    alpha = 0.5
  ) +
  labs(x = 'Odds Ratio', y = '', color = 'Model')

```

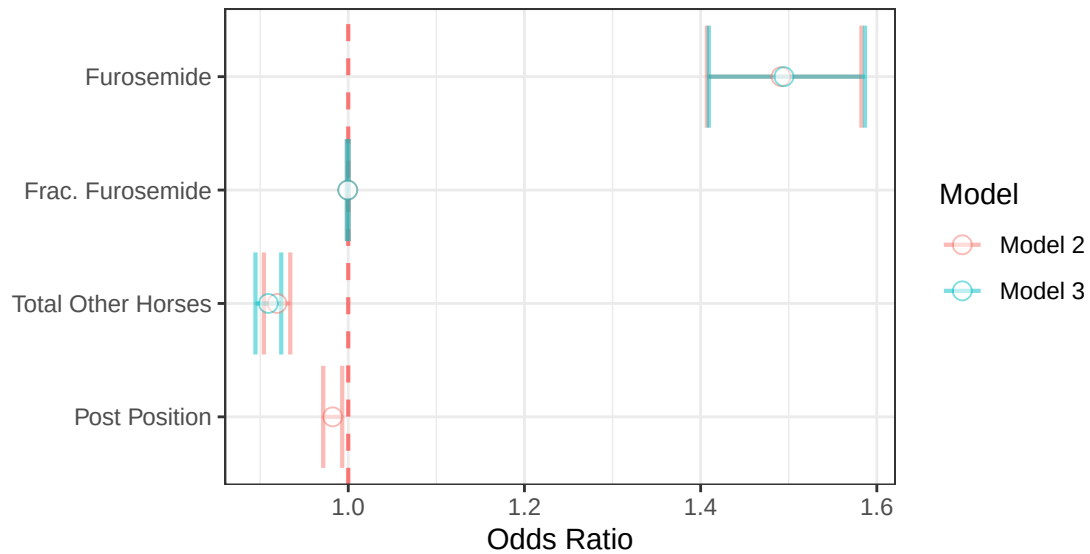


Figure A.2

```
# #| fig.cap: 'Predicted probabilities binned into 0.1-wide intervals.'
calibration_data |>
  filter(model != 'Cond. Logit') |>
  mutate(
    model = if_else(model == 'BHLR', 'Model 2', model),
    interval = cut(
      pred,
      breaks = seq(0, 1, by = .1),
      include.lowest = T
    )
  ) |>
  group_by(interval, model) |>
  summarize(
    int_pred = mean(pred),
    int_actual = mean(actual),
    se = sqrt((int_actual * (1 - int_actual)) / n()),
    n = n(),
    ci_lower = pmax(int_actual - 2 * se, 0),
    ci_upper = pmin(int_actual + 2 * se, 1)
  ) |>
  ggplot(aes(x = int_pred, y = int_actual, color = model)) +
  geom_point(alpha = 0.5, aes(size = n)) +
```

```

geom_errorbar(aes(ymin = ci_lower, ymax = ci_upper)) +
geom_smooth(se = FALSE, method = 'loess') +
geom_abline(slope = 1,
            intercept = 0,
            linetype = "dashed") +
labs(x = "Predicted probability",
     y = "Observed fraction",
     color = 'Model',
     size = 'Count') +
ylim(0, 1) +
theme(text = element_text(size = 15))

```

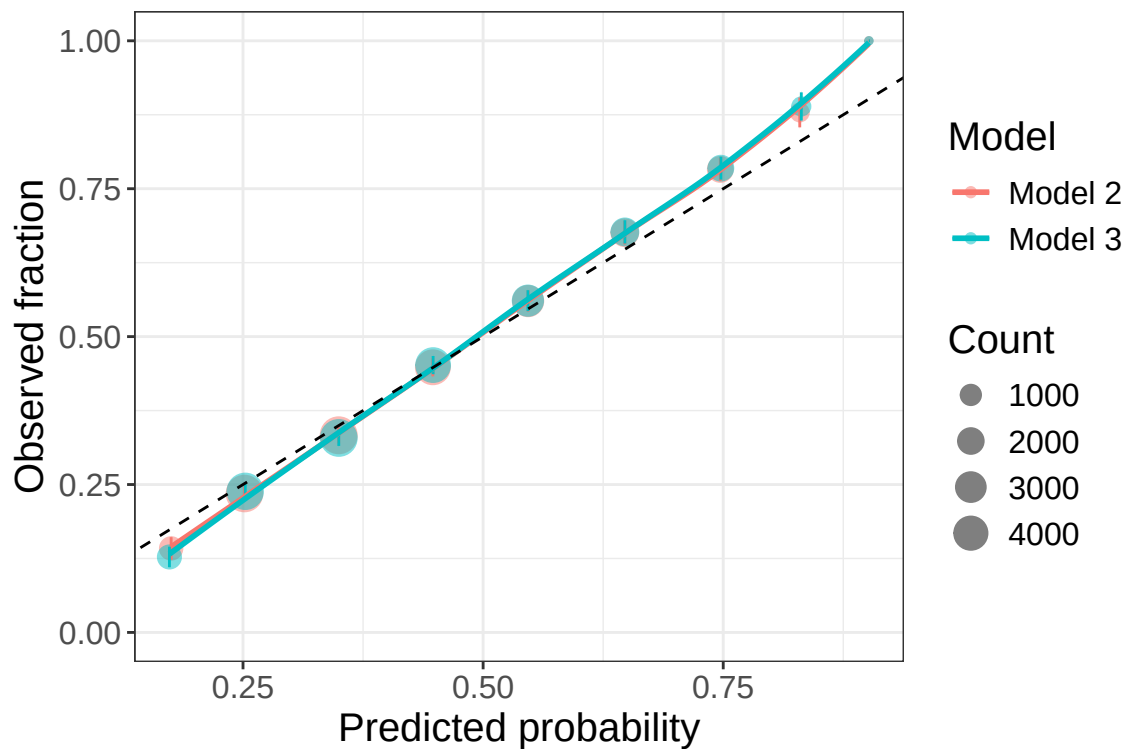


Table A.1

```

# BHLR model described in sec A.1 and eq (A.1)
summary(brms_fit)

```

Family: bernoulli

```

Links: mu = logit
Formula: in_the_money ~ med_description + (1 | registration_number) + (1 | track_id/post_pos.
Data: data_clean (Number of observations: 23109)
Draws: 4 chains, each with iter = 8000; warmup = 4000; thin = 1;
      total post-warmup draws = 16000

```

Multilevel Hyperparameters:

```

~jockey_id (Number of levels: 1935)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)      0.11      0.04      0.02      0.18 1.00      2316      2792

```

```

~registration_number (Number of levels: 8644)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)      0.27      0.07      0.11      0.38 1.01      854      786

```

```

~track_id (Number of levels: 88)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)      0.13      0.03      0.08      0.19 1.00      5877      7681

```

```

~track_id:post_position (Number of levels: 808)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)      0.08      0.04      0.01      0.16 1.00      2073      4087

```

Regression Coefficients:

```

      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
Intercept      2.24      0.11      2.02      2.45 1.00      18123
med_descriptionLasix      0.40      0.03      0.34      0.46 1.00      21255
splines::nsoddsdfEQ41     -2.09      0.08     -2.25     -1.93 1.00      15428
splines::nsoddsdfEQ42     -2.19      0.09     -2.38     -2.01 1.00      17220
splines::nsoddsdfEQ43     -4.38      0.19     -4.76     -4.01 1.00      14521
splines::nsoddsdfEQ44     -2.14      0.10     -2.33     -1.95 1.00      20540
frac_lasix       -0.00      0.00     -0.00      0.00 1.00      16649
total_other_horses     -0.10      0.01     -0.11     -0.08 1.00      24969

```

```

      Tail_ESS
Intercept      12449
med_descriptionLasix      11883
splines::nsoddsdfEQ41      12970
splines::nsoddsdfEQ42      12421
splines::nsoddsdfEQ43      12420
splines::nsoddsdfEQ44      11795
frac_lasix      11252
total_other_horses      12593

```

Draws were sampled using `sample(hmc)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat = 1`).

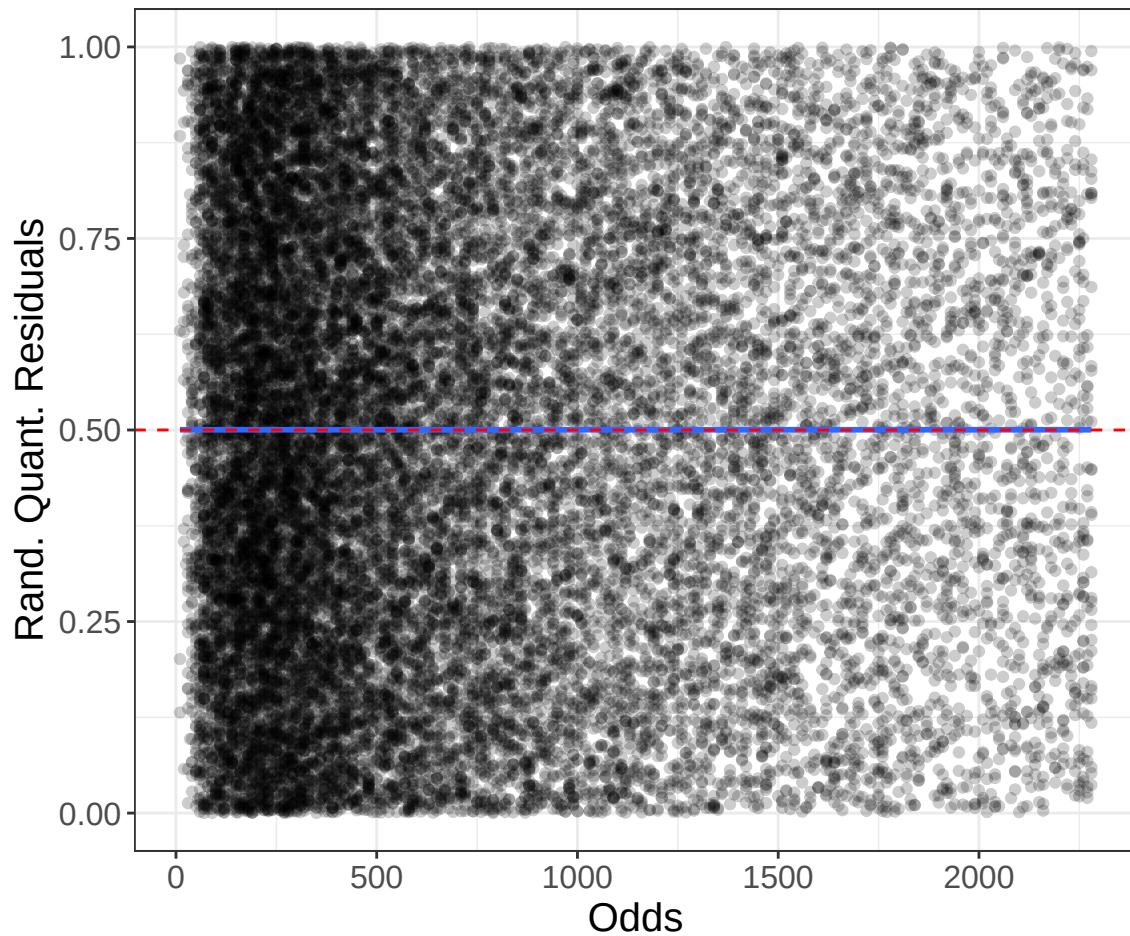
Section A.1.2 Randomized Quantile Residuals

Figure A.3

a

```
rqr_tbl <- data_clean |>
  mutate(brms_pred_apples = brms_pred_apples,
         cdf_bound = 1 - brms_pred_apples) |>
  rowwise() |>
  mutate(rq_resid = if_else(
    in_the_money == 0,
    runif(1, min = 0, max = cdf_bound),
    runif(1, min = cdf_bound, max = 1)
  )) |>
  ungroup()

ggplot(rqr_tbl, aes(x = odds, y = rq_resid)) +
  geom_point(alpha = 0.2) +
  geom_smooth(se = FALSE) +
  geom_hline(linetype = 'dashed',
            color = 'red',
            yintercept = .5) +
  ylim(0, 1) +
  theme(text = element_text(size = 15)) +
  labs(x = 'Odds', y = 'Rand. Quant. Residuals')
```



b

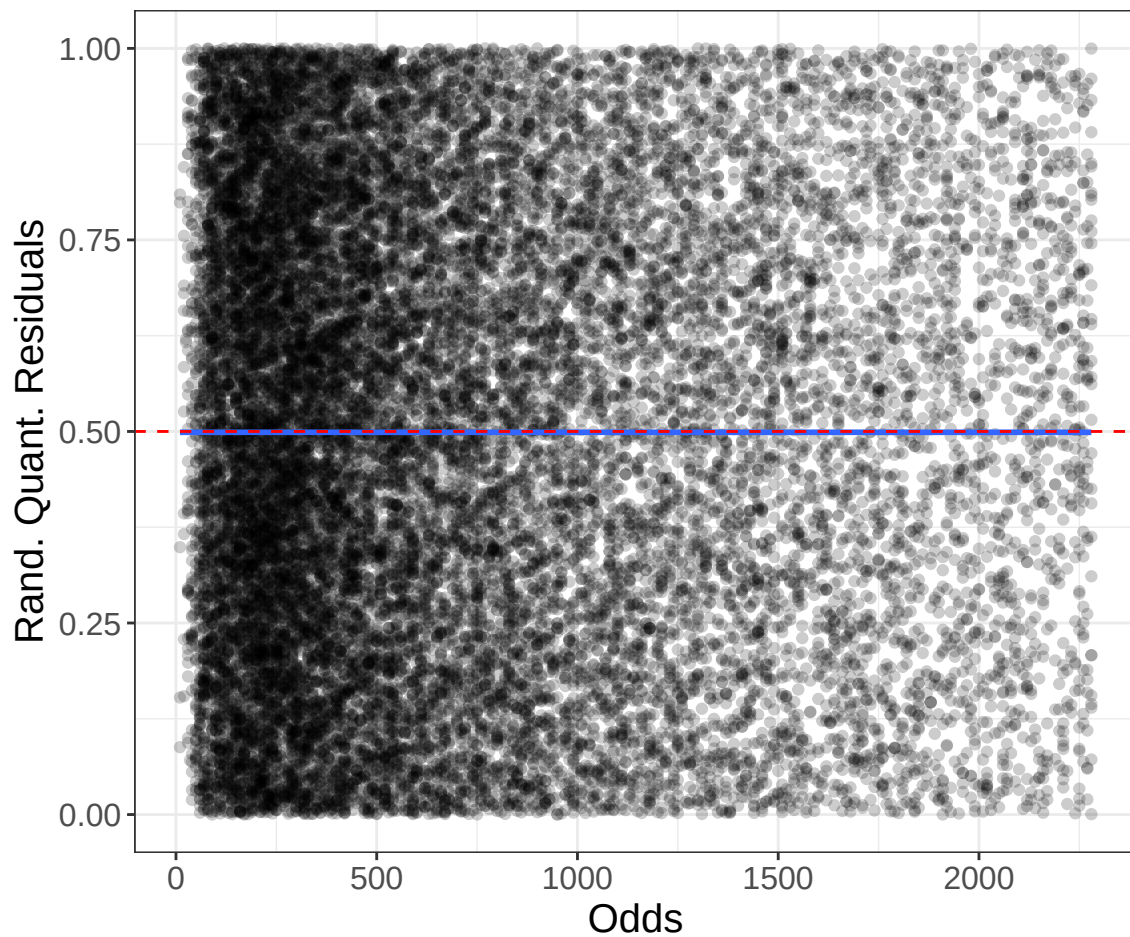
```
rqr_tbl <- data_clean |>
  mutate(brms_pred = brms_pred, cdf_bound = 1 - brms_pred) |>
  rowwise() |>
  mutate(rq_resid = if_else(
    in_the_money == 0,
    runif(1, min = 0, max = cdf_bound),
    runif(1, min = cdf_bound, max = 1)
  )) |>
  ungroup()

ggplot(rqr_tbl, aes(x = odds, y = rq_resid)) +
  geom_point(alpha = 0.2) +
```

```

geom_smooth(se = FALSE) +
geom_hline(linetype = 'dashed',
           color = 'red',
           yintercept = .5) +
ylim(0, 1) +
theme(text = element_text(size = 15)) +
labs(x = 'Odds', y = 'Rand. Quant. Residuals')

```

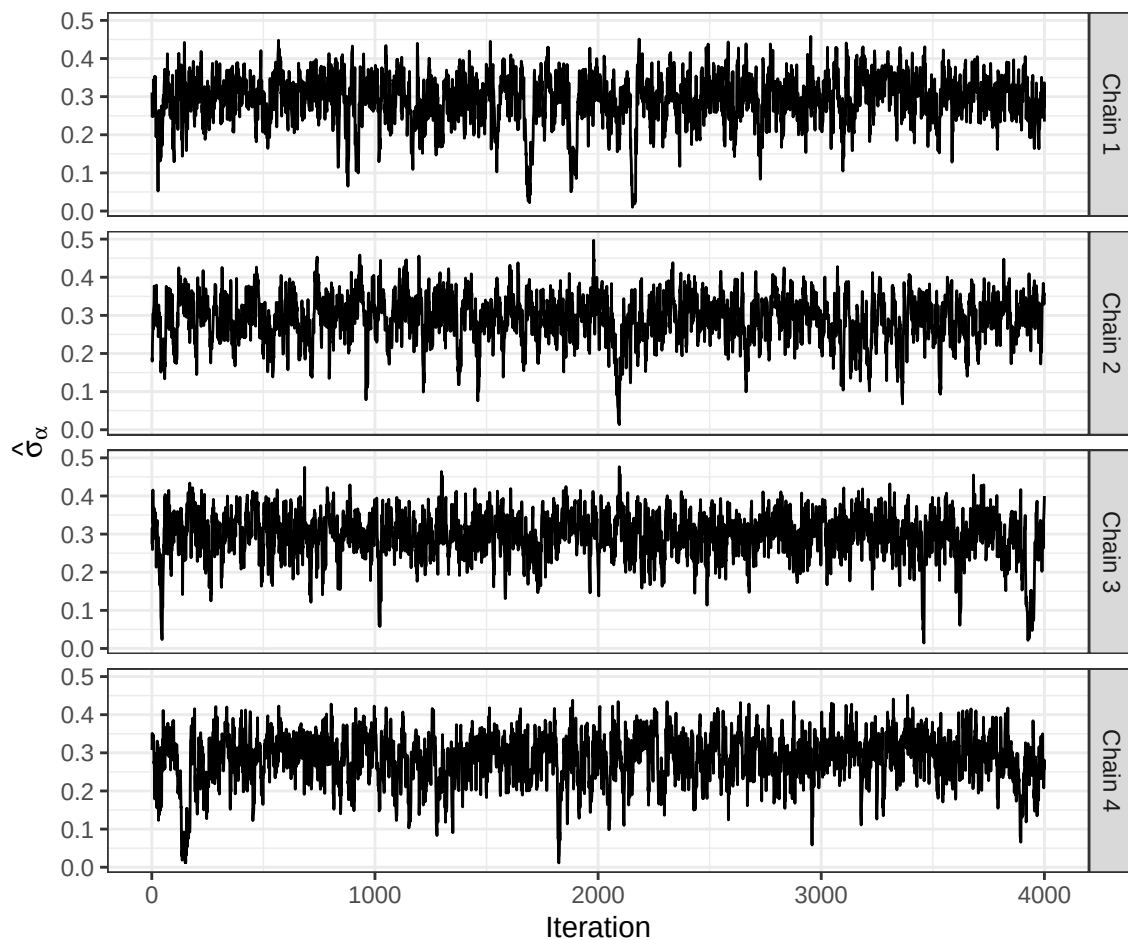


Section A.1.3 BHLR Trace Plots

Figure A.4

a

```
ggplot(  
  brms_apples_data |>  
    mutate(index = paste0('Chain ', index)),  
  aes(x = iter, y = sd_registration_number__Intercept)  
) +  
  geom_line() +  
  facet_grid(rows = vars(index)) +  
  labs(y = TeX('$\\hat{\\sigma}_\\alpha$'),  
       x = 'Iteration')
```



b

```
ggplot(brms_apples_data |>
  mutate(index = paste0('Chain ', index)),
  aes(x = iter, y = b_med_descriptionLasix)) +
  geom_line() +
  facet_grid(rows = vars(index)) +
  labs(y = TeX('$\\hat{\\tau}$'),
       x = 'Iteration')
```

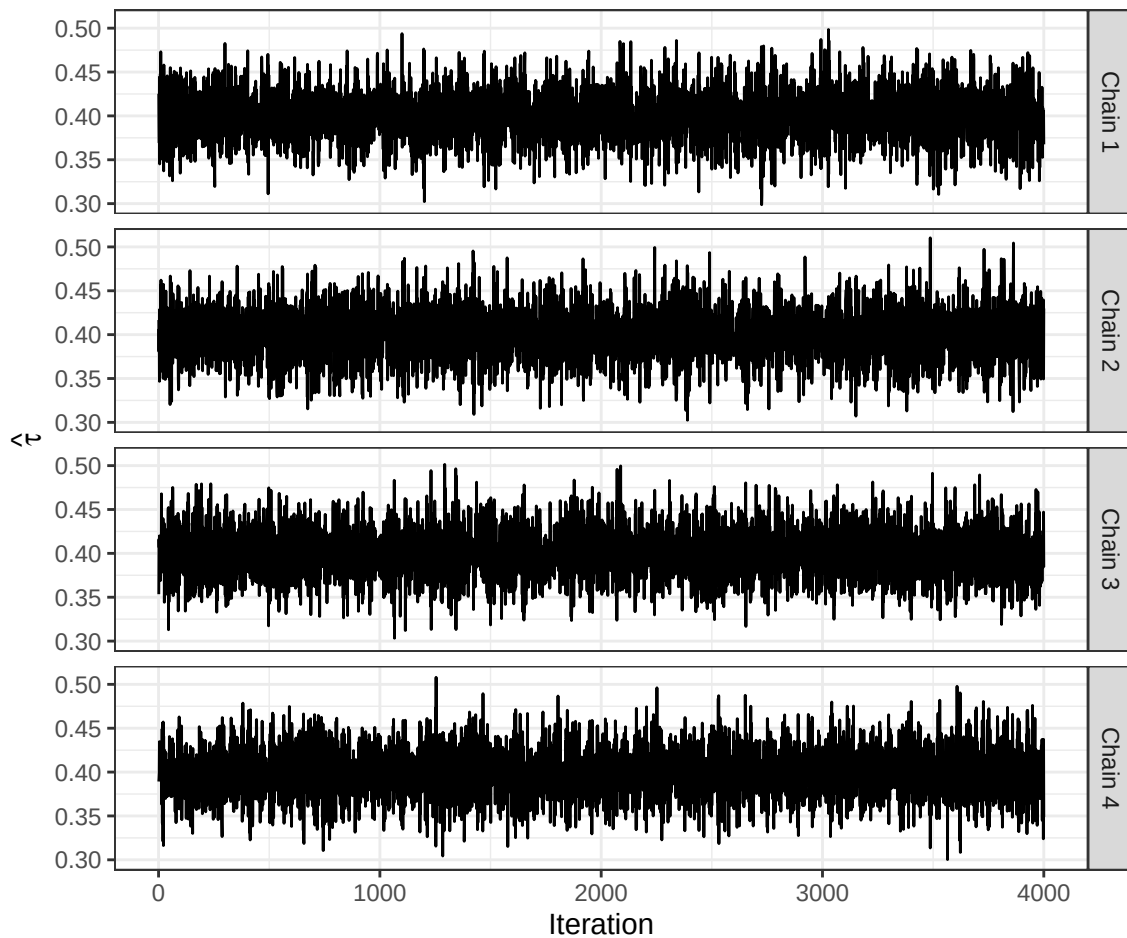
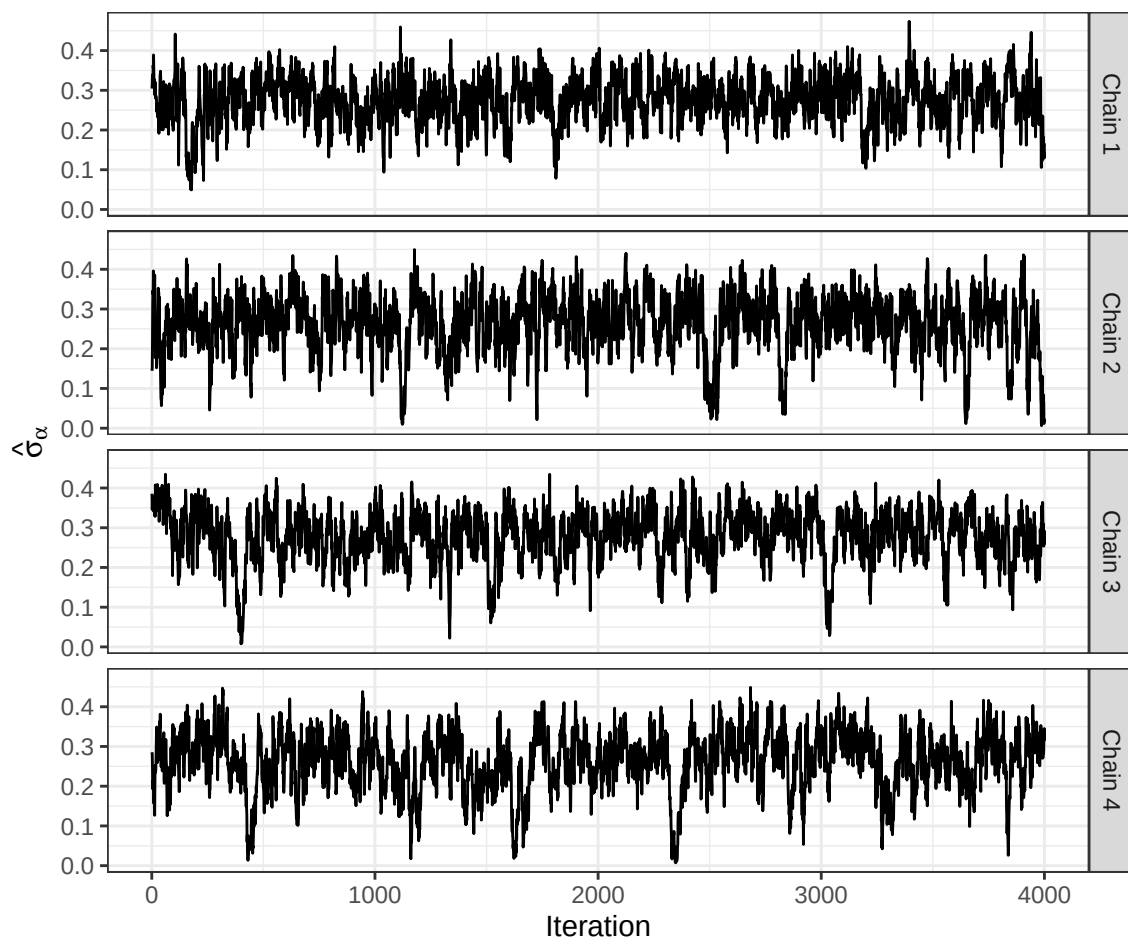


Figure A.5

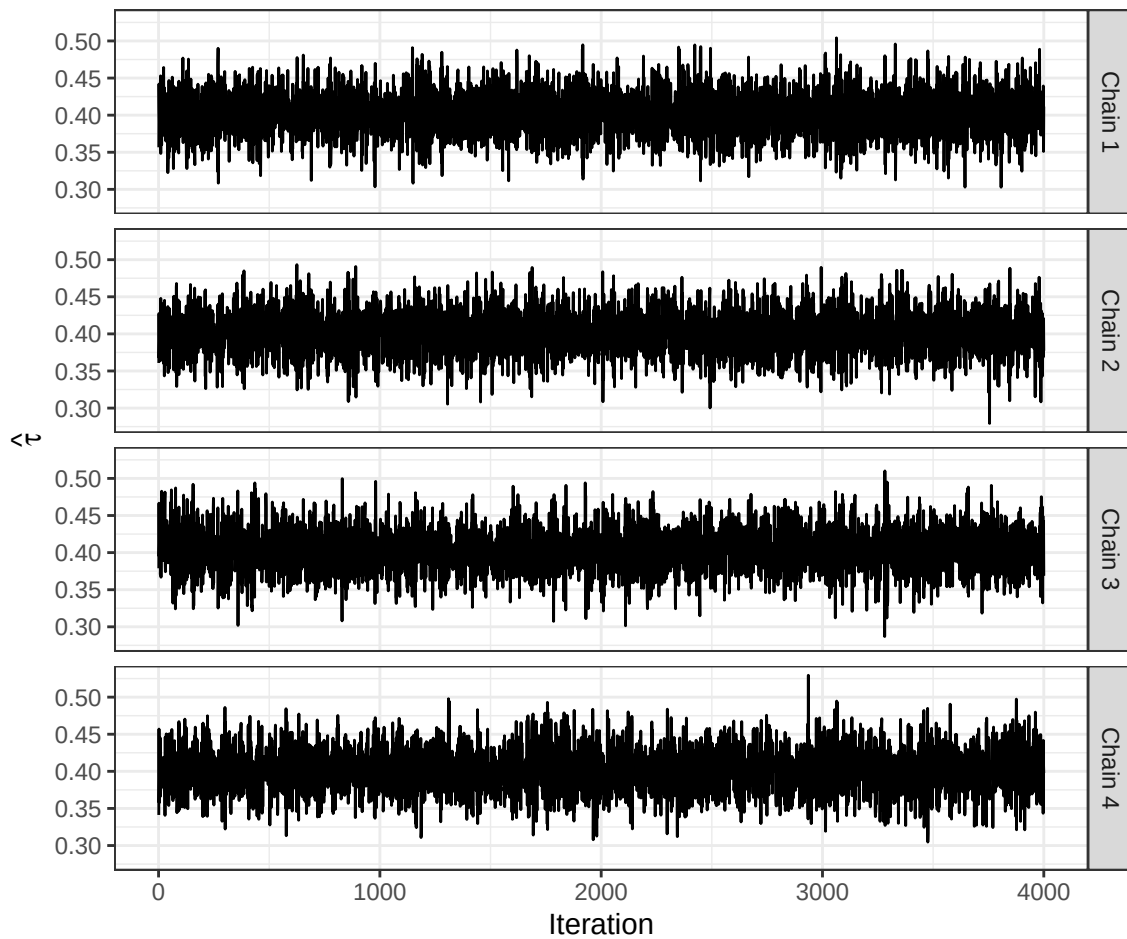
a

```
ggplot(
  brms_data |>
    mutate(index = paste0('Chain ', index)),
  aes(x = iter, y = sd_registration_number__Intercept)
) +
  geom_line() +
  facet_grid(rows = vars(index)) +
  labs(y = TeX('$\\hat{\\sigma}_\\alpha$'),
       x = 'Iteration')
```



b

```
ggplot(brms_data |>
  mutate(index = paste0('Chain ', index)),
  aes(x = iter, y = b_med_descriptionLasix)) +
  geom_line() +
  facet_grid(rows = vars(index)) +
  labs(y = TeX('$\\hat{\\tau}$'),
       x = 'Iteration')
```



...